Course : AI Assistant Coding

Instructor : Sallauddin Mohammed

Title : Car Rental System

Team Details : Pannati Samanvith (2303A52090)

Introduction :

The Car Rental System is a full-stack web application that allows users to:

- Browse available cars
- Book cars with drivers
- View booking history

The system also includes an **Admin Panel** to manage:

- Cars
- Drivers
- Bookings

Objective :

Build a full-stack application using MERN stack

- Provide seamless car booking experience
- Implement authentication using JWT
- Manage resources (cars, drivers) efficiently
- Apply AI-assisted development for faster coding

Stakeholders :

- **Users:** Book cars
- **Admin:** Manage cars, drivers, bookings
- **Developers:** Maintain system
- **Business Owner:** Revenue from bookings

Technologies Used :

Frontend:

- React.js
- CSS (inline styling)

Backend:

- Node.js
- Express.js

Database:

- MongoDB

Tools:

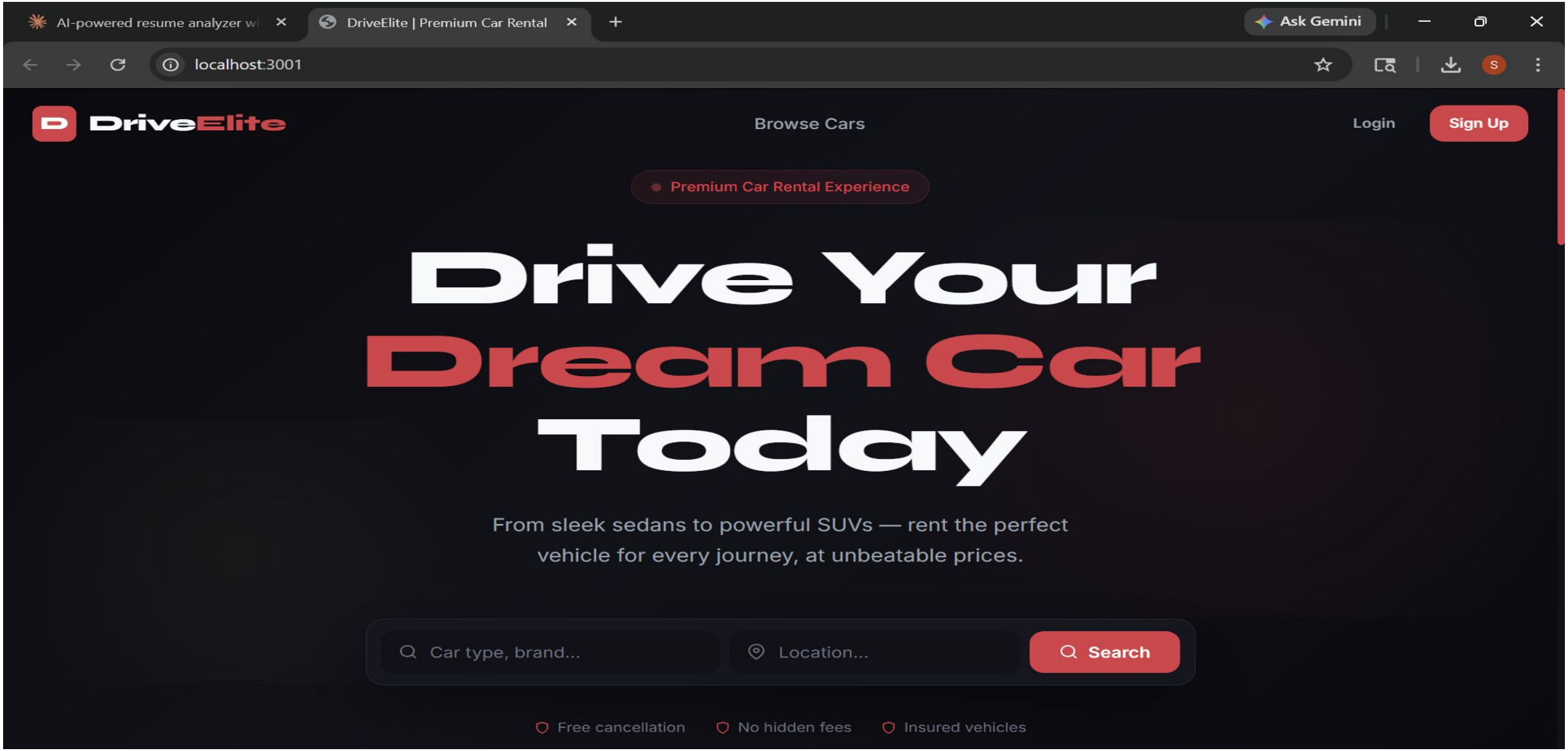
- GitHub
- VS Code

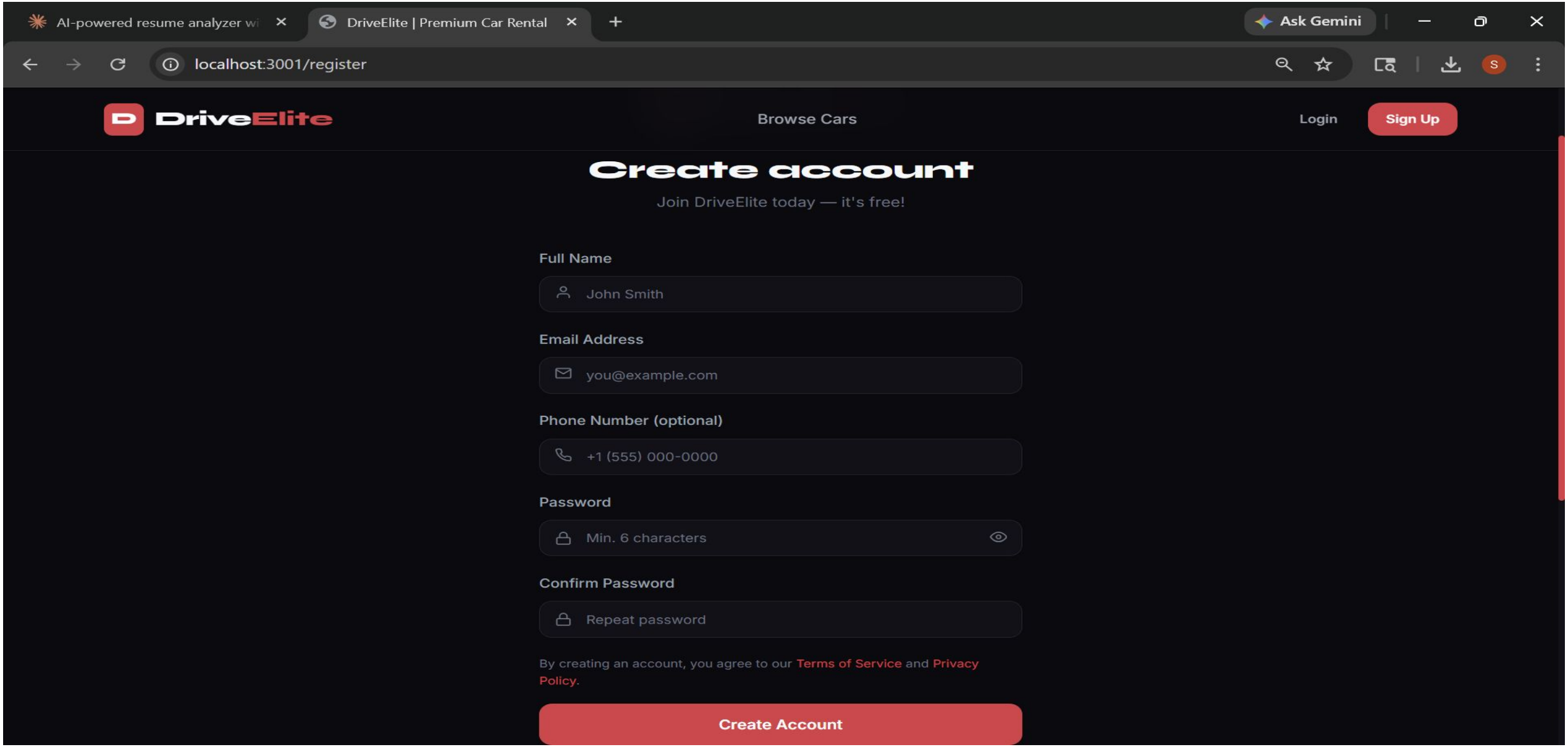
Authentication:

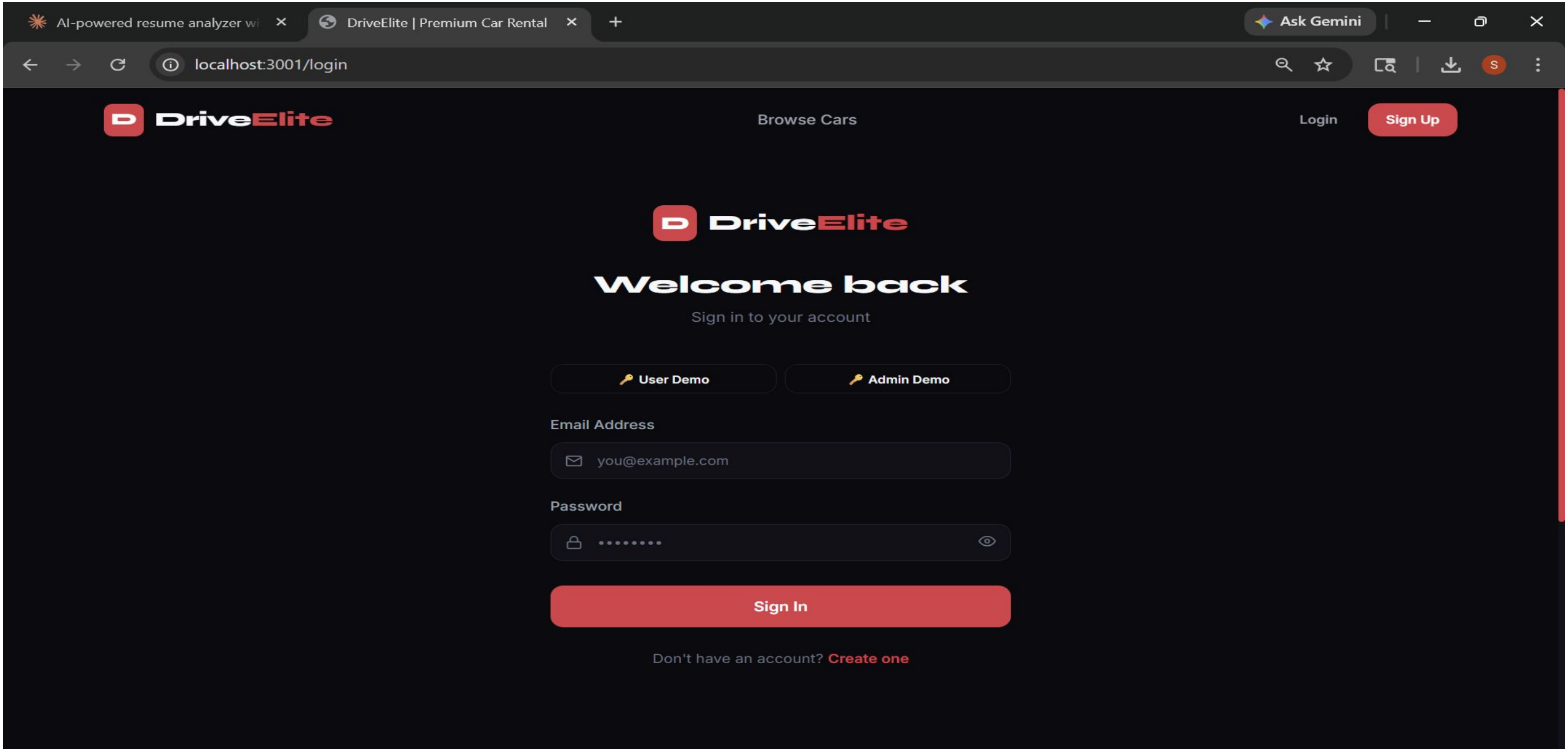
- JWT (JSON Web Token)

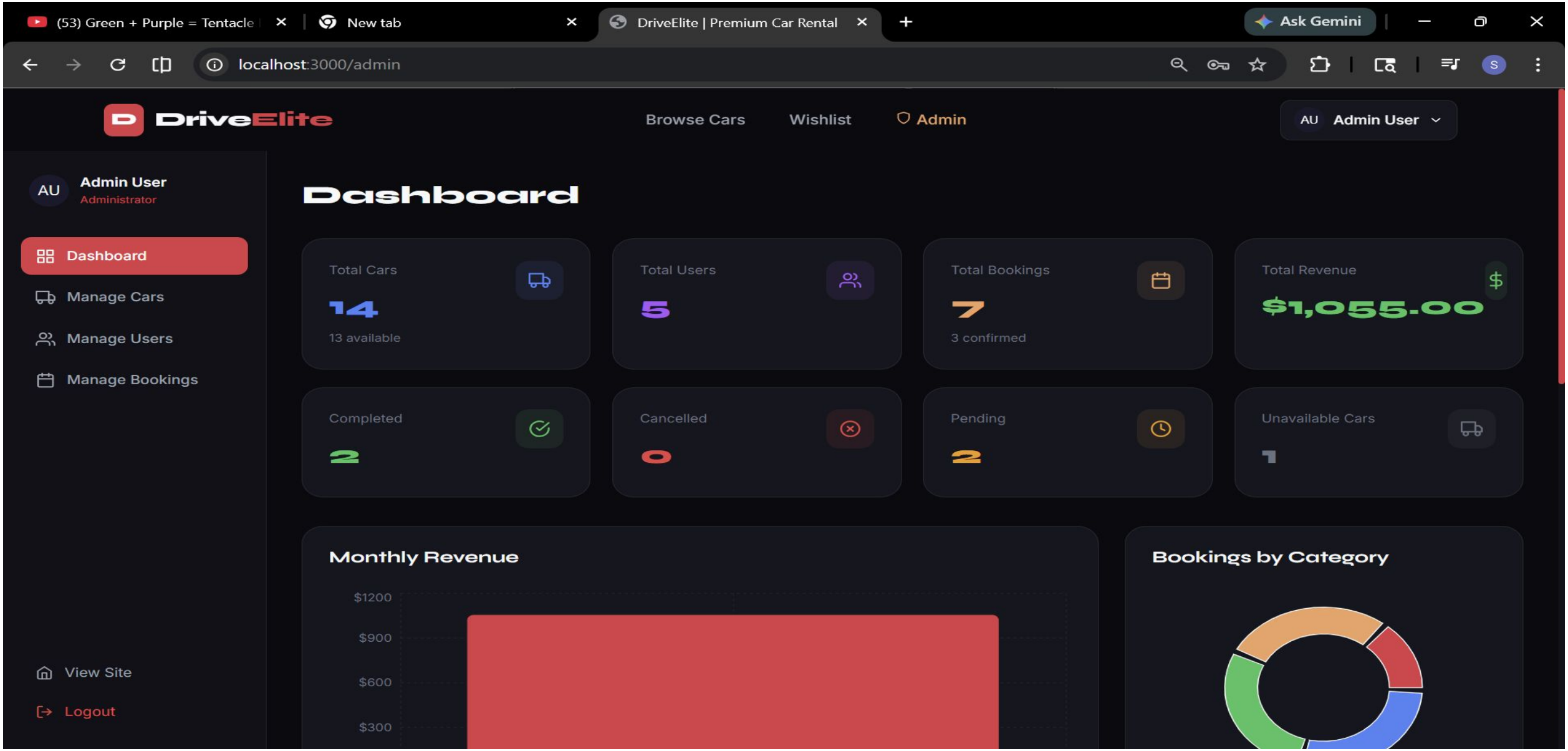
AI Prompts Used :

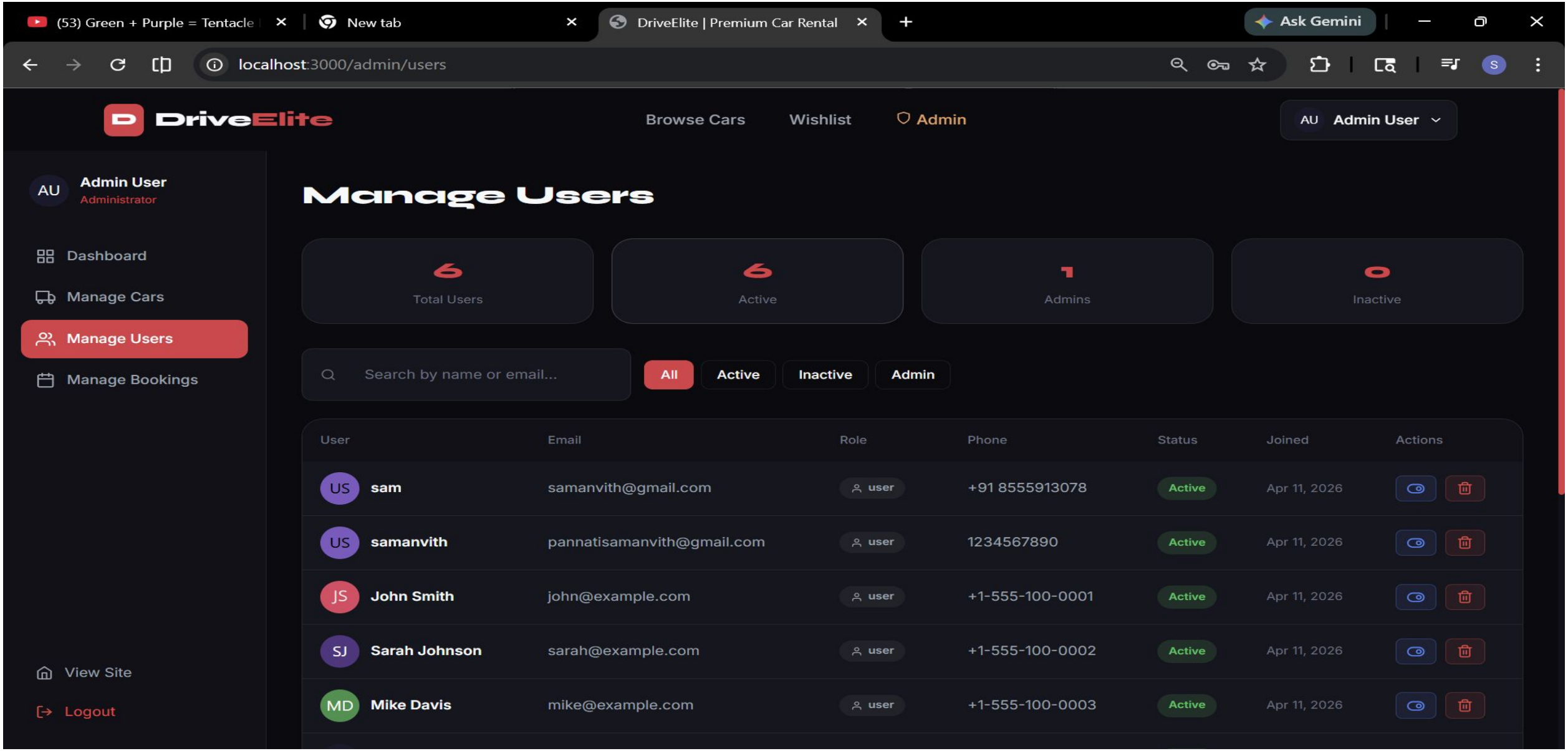
- Create a React car booking page with form and API integration”
- “Fix 401 unauthorized JWT error in MERN app”
- “Improve UI for car cards with premium styling”
- “Create Admin dashboard with CRUD operations”
- “Filter available drivers in booking system”

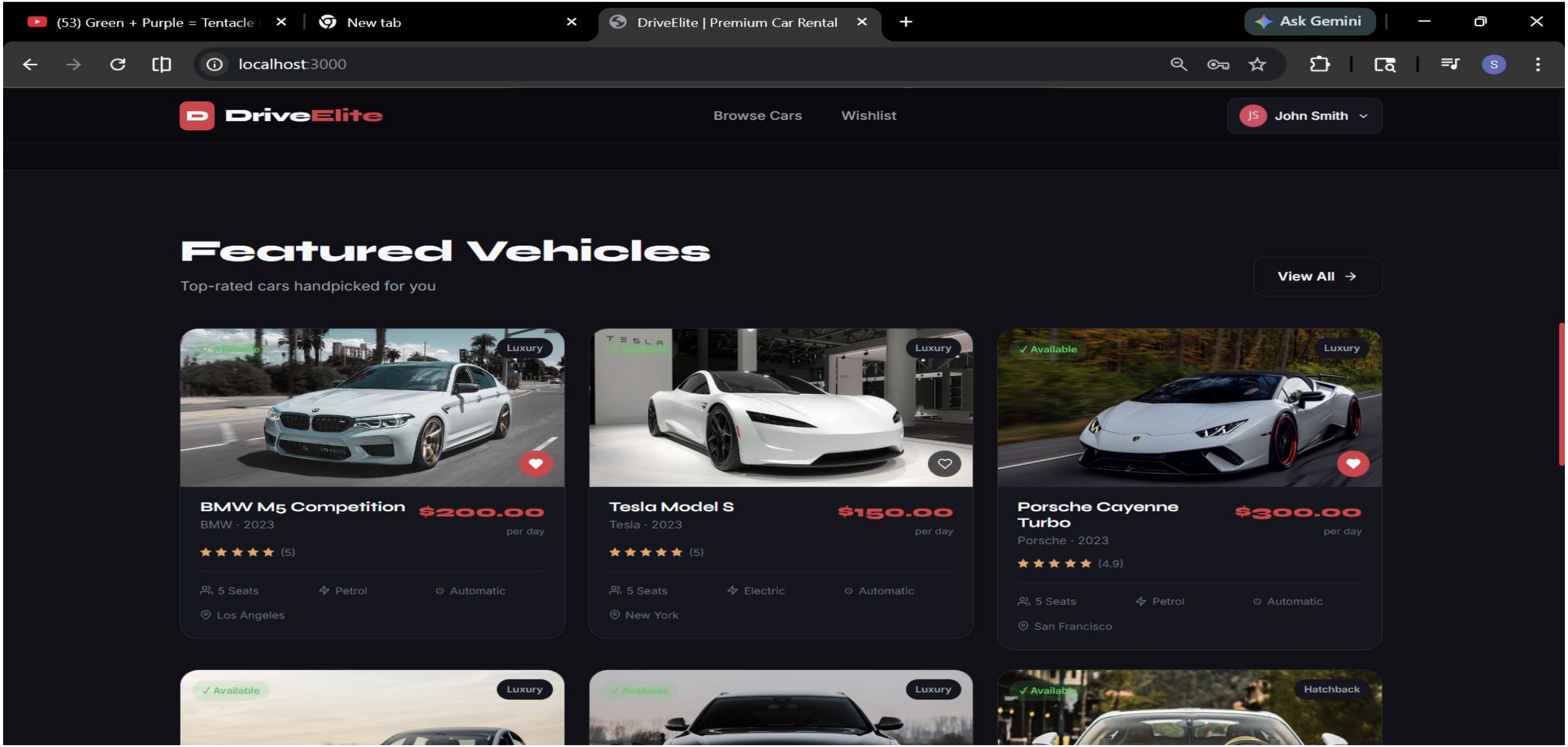

AI-powered resume analyzer wi
DriveElite | Premium Car Rental
Ask Gemini
localhost:3001
DriveElite
Browse Cars
Login
Sign Up
Premium Car Rental Experience
Drive Your
Dream Car
Today
From sleek sedans to powerful SUVs — rent the perfect vehicle for every journey, at unbeatable prices.
Car type, brand...
Location...
Search
Free cancellation
No hidden fees
Insured vehicles


AI-powered resume analyzer wi
DriveElite | Premium Car Rental
Ask Gemini
localhost:3001/register
DriveElite
Browse Cars
Login
Sign Up
Create account
Join DriveElite today — it's free!
Full Name
John Smith
Email Address
you@example.com
Phone Number (optional)
+1 (555) 000-0000
Password
Min. 6 characters
Confirm Password
Repeat password
By creating an account, you agree to our Terms of Service and Privacy Policy.
Create Account


AI-powered resume analyzer wi
DriveElite | Premium Car Rental
Ask Gemini
localhost:3001/login
DriveElite
Browse Cars
Login
Sign Up
DriveElite
Welcome back
Sign in to your account
User Demo
Admin Demo
Email Address
you@example.com
Password
Sign In
Don't have an account? Create one


(53) Green + Purple = Tentacle
New tab
DriveElite | Premium Car Rental
Ask Gemini
localhost:3000/admin
DriveElite
Browse Cars
Wishlist
Admin
AU Admin User
AU Admin User
Administrator
Dashboard
Manage Cars
Manage Users
Manage Bookings
View Site
Logout
Dashboard
Total Cars
14
13 available
Total Users
5
Total Bookings
7
3 confirmed
Total Revenue
$1,055.00
Completed
2
Cancelled
0
Pending
2
Unavailable Cars
1
Monthly Revenue
$1200
$900
$600
$300
Bookings by Category


(53) Green + Purple = Tentacle
New tab
DriveElite | Premium Car Rental
Ask Gemini
localhost:3000/admin/users
DriveElite
Browse Cars
Wishlist
Admin
AU Admin User
AU Admin User
Administrator
Dashboard
Manage Cars
Manage Users
Manage Bookings
View Site
Logout
Manage Users
6 Total Users
6 Active
1 Admins
0 Inactive
Search by name or email...
All
Active
Inactive
Admin
User
Email
Role
Phone
Status
Joined
Actions
US sam samanvith@gmail.com user +91 8555913078 Active Apr 11, 2026
US samanvith pannatisamanvith@gmail.com user 1234567890 Active Apr 11, 2026
JS John Smith john@example.com user +1-555-100-0001 Active Apr 11, 2026
SJ Sarah Johnson sarah@example.com user +1-555-100-0002 Active Apr 11, 2026
MD Mike Davis mike@example.com user +1-555-100-0003 Active Apr 11, 2026


(53) Green + Purple = Tentacle
New tab
DriveElite | Premium Car Rental
Ask Gemini
localhost:3000
DriveElite
Browse Cars
Wishlist
JS
John Smith
Featured Vehicles
Top-rated cars handpicked for you
View All
Luxury
BMW M5 Competition
BMW · 2023
$200.00
per day
(5)
5 Seats
Petrol
Automatic
Los Angeles
Luxury
Tesla Model S
Tesla · 2023
$150.00
per day
(5)
5 Seats
Electric
Automatic
New York
✓ Available
Luxury
Porsche Cayenne Turbo
Porsche · 2023
$300.00
per day
(4.9)
5 Seats
Petrol
Automatic
San Francisco
✓ Available
Luxury
✓ Available
Luxury
✓ Available
Hatchback

(53) Green + Purple = Tentacle | New tab | DriveElite | Premium Car Rental | Ask Gemini

localhost:3000/cars

DriveElite

Browse Cars Wishlist JS John Smith

Browse Cars

14 vehicles available

Filters

× Clear All

Search

Car name, brand...

Location

City or airport...

Category

All Categories

Body Type

All Types

Fuel Type

All Fuel Types

Sort: Default

kia
✓ Available Economy
kia
malik · 2026
$799.00 per day
(0)
5 Seats Petrol Automatic
hnk

swift
✓ Available Economy
swift
maruthi · 2026
$899.00 per day
(0)
5 Seats Petrol Automatic
hnk

SUV
Jeep Wrangler Rubicon
Jeep · 2023
$110.00 per day
(4.5)
5 Seats Petrol Manual
Denver

✓ Available Premium

Toyota Camry Hybrid
✓ Available Economy

✓ Available Hatchback

localhost:3000/cars/69d9f0893ca05a9701087538

Proof Of AI Prompts :

File Edit Selection View Go Run ... car-rental-system

EXPLORER

CAR-RENTAL-SY...
- .github\workflows
  - ci.yml
- client
  - node_modules
  - public
  - src
    - api
    - assets
    - components
      - BookingForm.jsx
      - CarCard.jsx
      - DriverSelect.jsx
      - Navbar.jsx
      - ProtectedRoute.jsx
    - pages
    - services
    - App.css
    - App.jsx
    - index.css
    - main.jsx
  - .gitignore
  - eslint.config.js
  - index.html
  - package-lock.json
  - package.json
  - README.md

OUTLINE
TIMELINE
MYSQL

ulatePrice.js | calculateDistance.js | bookingController.js | Dashboard.jsx | BookingForm.jsx | api.js ...\

client > src > components > BookingForm.jsx > default

You, 5 days ago | 1 author (You)

```
// Create a React component for a car booking form that allows users to enter trip details
// such as pickup location, drop location, start date, and end date. Use useState to manage
// form inputs and selected driver ID, and use useEffect to fetch available drivers from the API
// when the component loads. Integrate a DriverSelect component to display and choose a driver.
import { useEffect, useState } from "react";
import axios from "../api/api";
import DriverSelect from "./DriverSelect";
import { useNavigate } from "react-router-dom";

const BookingForm = ({ carId }) => {
  const navigate = useNavigate();

  const [drivers, setDrivers] = useState([]);
  const [driverId, setDriverId] = useState("");
  const [car, setCar] = useState(null); // ✅ NEW

  const [form, setForm] = useState({
    pickupLocation: "",
    dropLocation: "",
    startDate: "",
    endDate: ""
  });

  const [distance, setDistance] = useState(0);
  const [price, setPrice] = useState(0);
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS | GITLENS | POSTMAN CONSOLE | node - client

```
PS C:\Users\dasar\car-rental-system\client> npm run dev
  ➜  Local:   http://localhost:5173/
  ➜  Network: use --host to expose
  ➜  press h + enter to show help
```

```
PS C:\Users\dasar\car-rental-system\server> npm run dev
MongoDB Connected
Server running on port 5000
```

File Edit Selection View Go Run car-rental-system

EXPLORER

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- .github\workflows
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      - ProtectedRoute.jsx
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    - services
    - App.css
    - App.jsx
    - index.css
    - main.jsx
  - .gitignore
  - eslint.config.js
  - index.html
  - package-lock.json
  - package.json
  - README.md
- OUTLINE
- TIMELINE
- MYSQL

CarCard.jsx | Navbar.jsx | DriverSelect.jsx | ProtectedRoute.jsx | index.css | authMiddleware.js | User.js

client > src > components > CarCard.jsx > default

```jsx
// Create a reusable React component to display car details in a card format.
// Accept a car object as a prop and safely handle cases where the car data may be undefined.
// Show the car name, availability status, base rent, and price per kilometer with clean UI styling.
import { useNavigate } from "react-router-dom";

const CarCard = ({ car }) => {
  const navigate = useNavigate();

  if (!car) return null;

  const handleBooking = () => {
    if (car._id) {
      navigate(`/booking/${car._id}`);
    }
  };

  return (
    <div style={styles.card}>

      {/* HEADER */}
      <div style={styles.top}>
        <div>
          <h3 style={styles.name}>{car?.name || "Unnamed Car"}</h3>
          <p style={styles.category}>
            {car?.category?.toUpperCase() || "ECONOMY"}
          </p>
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS POSTMAN CONSOLE

node - client

```
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File Edit Selection View Go Run car-rental-system

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- OUTLINE
- TIMELINE
- MYSQL

CarCard.jsx | Navbar.jsx | DriverSelect.jsx 1 | ProtectedRoute.jsx | index.css | authMiddleware.j

client > src > components > DriverSelect.jsx > default

You, 5 days ago | 1 author (You)

```
// Create a reusable React component for selecting a driver from a dropdown list.
// Accept a list of drivers and a setter function as props to update the selected driver ID.
// Render a select input with a default "Select Driver" option.
// Create a reusable React component for selecting a driver from a dropdown list.
// Accept a list of drivers and a setter function as props to update the selected driver ID.
// Render a select input with a default "Select Driver" option.
import { useState, useEffect } from "react";
import axios from "../api/api";

const DriverSelect = ({ setDriverId }) => {
  const [drivers, setDrivers] = useState([]);
  const [selected, setSelected] = useState("");

  useEffect(() => {
    fetchDrivers();
  }, []);

  const fetchDrivers = async () => {
    try {
      const res = await axios.get("/drivers");

      // ✅ only available drivers
      const available = res.data.filter((d) => d.available === true);

      setDrivers(available);
```

PROBLEMS 2 | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS | GITLENS | POSTMAN CONSOLE | node - client

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PS C:\Users\dasar\car-rental-system\client> npm run dev
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  ➜  press h + enter to show help
```

```
PS C:\Users\dasar\car-rental-system\server> npm run dev
MongoDB Connected
Server running on port 5000
```

References

- React Documentation
- MongoDB Docs
- Node.js Docs
- Express.js Docs
- ChatGPT (OpenAI)
- Stack Overflow

Github Link

- https://github.com/samanvith7/Car-Rental-Booking-System

THANK YOU!